Polar Equations and Graphs

Recognizing polar graphs

1 Identify the graph of each polar equation:

a $r = 3$

b $\theta = \frac{\pi}{6}$

c $r = 2 + 2\cos\theta$

d $r = 4\sin 3\theta$

2 For the limaçon $r = 1 + 2\cos\theta$:

a Explain how you know it has an inner loop.

b Find the values of θ in $[0, 2\pi)$ at which the curve passes through the pole.

3 For the rose $r = 5\sin 2\theta$:

a State the number of petals.

b State the maximum value of $|r|$ and the smallest positive θ at which it occurs.

4 Determine whether the graph of $r = 2 + \sin\theta$ passes through the pole, and state the type of limaçon.

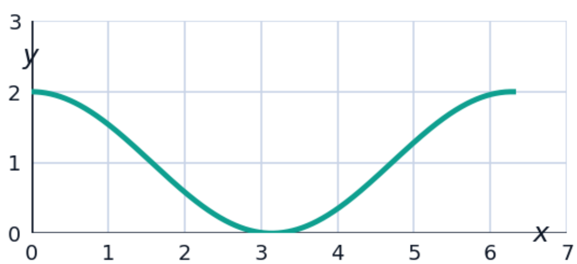
Symmetry tests

5 Test $r = 3\cos 2\theta$ for symmetry about the polar axis (replace θ with $-\theta$) and about the pole (replace r with $-r$).

6 Test $r = 2 + 2\sin\theta$ for symmetry about the line $\theta = \frac{\pi}{2}$ (replace θ with $\pi - \theta$).

Graphing from a table of values

7 For the cardioid $r = 1 + \cos\theta$, complete the table of r-values: $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.



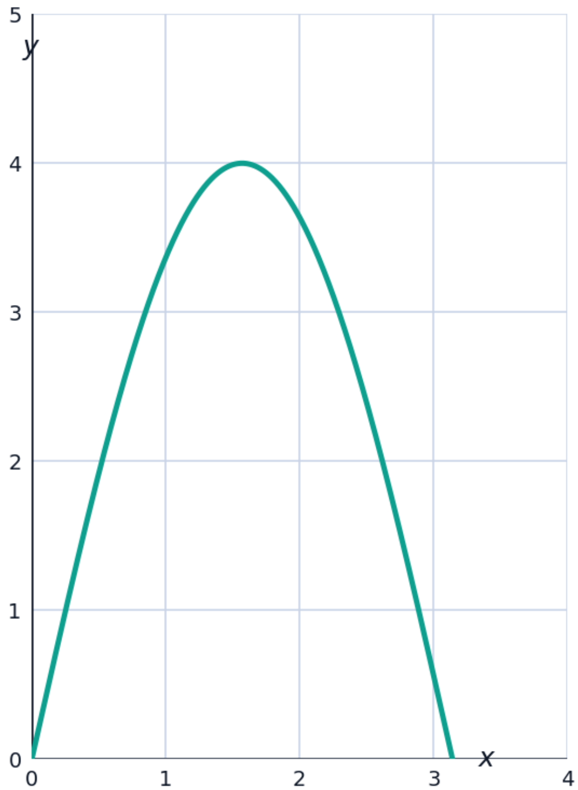
a $r(0)$

b $r\left(\frac{\pi}{2}\right)$

c $r(\pi)$

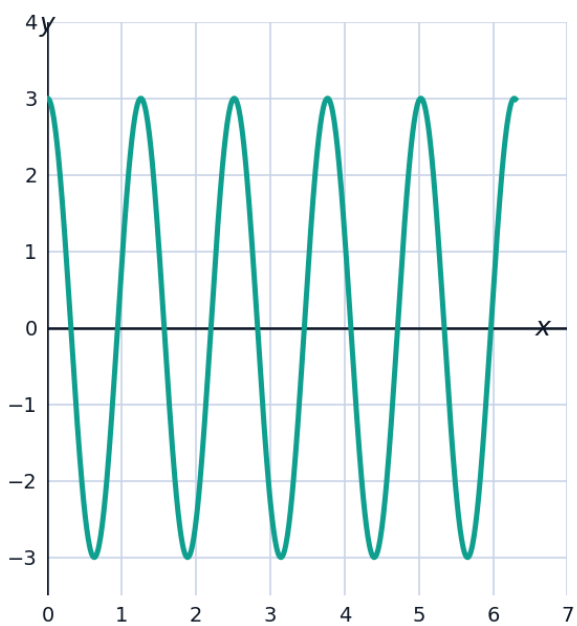
d $r\left(\frac{3\pi}{2}\right)$

8 For the circle $r = 4\sin\theta$, find the value(s) of θ in $[0, \pi)$ where r reaches its maximum, and state that maximum value.



Rose curves in depth

9 For the rose $r = 3\cos 5\theta$, state the number of petals and the length of each petal (maximum $|r|$).



10 Explain, using the general rule for rose curves $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$, why n odd gives n petals but n even gives $2n$ petals.

Identifying limaçon subtypes

11 Classify each limaçon $r = a + b\cos\theta$ as having an inner loop, being dimpled, or being convex (roughly oval, no dimple), based on the ratio $\frac{a}{b}$:

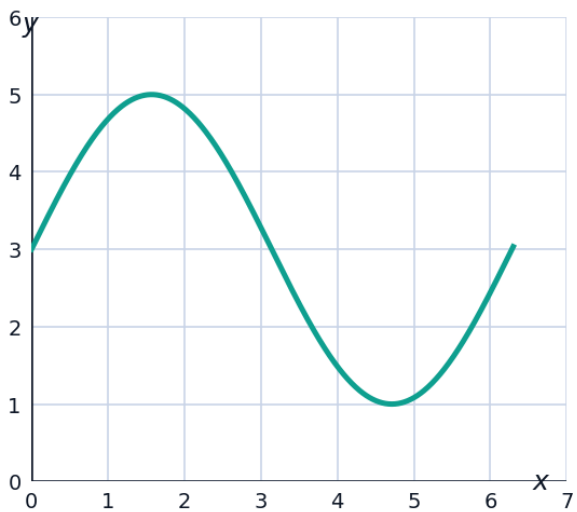
a $r = 1 + 3\cos\theta$

b $r = 3 + 2\cos\theta$

c $r = 4 + \cos\theta$

Finding maximum and minimum r-values

12 For the limaçon $r = 3 + 2\sin\theta$, find the maximum and minimum values of r, and the angles at which they occur.



Comparing sine- and cosine-based polar curves

13 Explain how the graph of $r = 3 + 3\sin\theta$ compares to $r = 3 + 3\cos\theta$ (same cardioid shape, different orientation).

Sketching from key features

14 Without fully graphing, list the key features you would compute first to sketch $r = 2 - 4\cos\theta$ (max/min r, pole crossings, symmetry).

A capstone graphing problem

15 For the rose $r = 4\cos(3\theta)$, find: (a) the number of petals, (b) the maximum value of r, (c) the value(s) of $\theta \in [0, \pi)$ where the first petal reaches its maximum length, and (d) the angles where the curve passes through the pole in $[0, \pi)$.

Distinguishing curve families by equation form alone

16 Without graphing, state whether each polar equation is a circle, a line, a rose, or a limaçon, based purely on its algebraic form: $r = 7$, $r = 5\sin(4\theta)$, $\theta = \frac{2\pi}{5}$, $r = 2 - 2\cos\theta$.